

LAST NAME: _____

Practice Quiz 02D — MATH 2055 — Fall 2026

Practice for Quiz 02

The following is closed book and closed note. Don't assume the grader will know what you mean. Use full sentences and provide complete and correct solutions.

Revised using GPT-6 Astra in the Codex harness.

1. A student claims that renaming the bound variable x as y changes

$$\forall x \in \mathbf{R} (x = y) \quad \text{into} \quad \forall y \in \mathbf{R} (y = y)$$

without changing the meaning.

- (a) Identify the free and bound variables in each formula, and explain what is wrong with the student's claim.

- (b) Correctly rename the bound variable of the first formula as t .